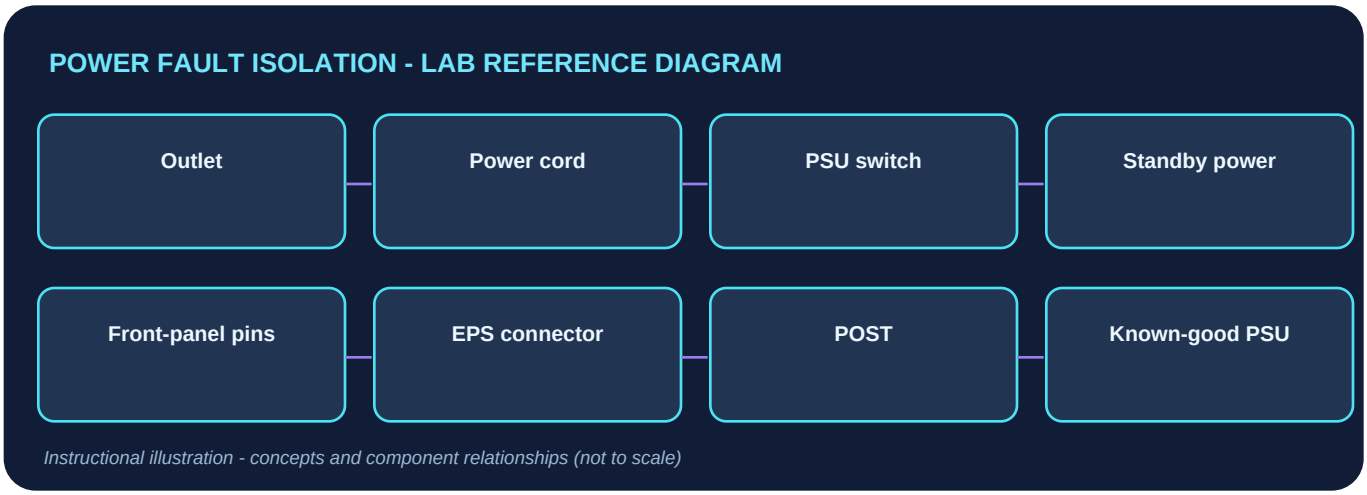
4.2.2 Troubleshoot System Power

STUDENT STUDY GUIDE + VOCABULARY + VISUAL REFERENCE

Learning goal

Find causes of power and POST failures.

Reference image



Use the labeled illustration to identify the key components, commands, or checkpoints you will encounter in this lab.

Prepare for the simulation

1. Verify wall power and external cord.
2. Inspect PSU switch and power connections.
3. Observe standby LEDs, fans and POST indicators.
4. Substitute known-good components where safe.

Troubleshooting and safe practice

• Completely dead: trace power source to PSU input.

• Fans but no POST: inspect RAM and CPU power.

4.2.2 | VOCABULARY & REVIEW

Essential vocabulary

Term	Meaning / recognition cue
Outlet	A key term in power fault isolation; identify its purpose, where it is used, and how to verify it.
Power cord	Electrical power characteristic or component; assess safely with rated tools and correct connectors.
PSU switch	A key term in power fault isolation; identify its purpose, where it is used, and how to verify it.
Standby power	Electrical power characteristic or component; assess safely with rated tools and correct connectors.
Front-panel pins	A key term in power fault isolation; identify its purpose, where it is used, and how to verify it.
EPS connector	Physical interface or connection point; verify its intended function and compatibility before use.
POST	Power-on self-test; firmware checks essential hardware during startup.
Known-good PSU	A key term in power fault isolation; identify its purpose, where it is used, and how to verify it.

Knowledge check - answer in your own words

1. Identify two components or commands in the reference diagram and explain what each does.

2. What should you verify before changing or installing hardware, settings, or files in this lab?

3. Choose one listed failure symptom. What is your first safe diagnostic step?

4. How will you prove the task is complete, rather than merely assuming success?

Lab readiness checklist

[ ] I can identify the labeled reference items. [ ] I understand the safety precautions.

[ ] I can explain the vocabulary. [ ] I can verify and document successful results.

Student: _____ Date: _____